

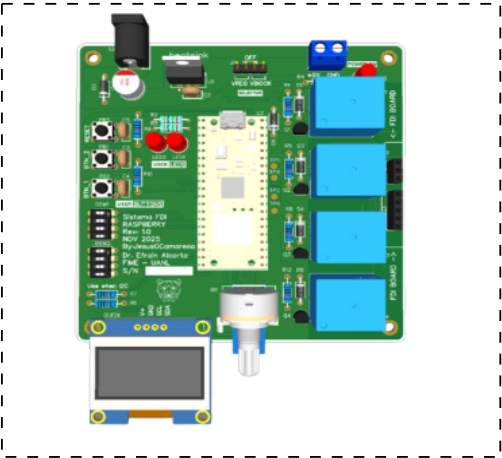
COVER PAGE

DATE: 15/11/2025
V1.0

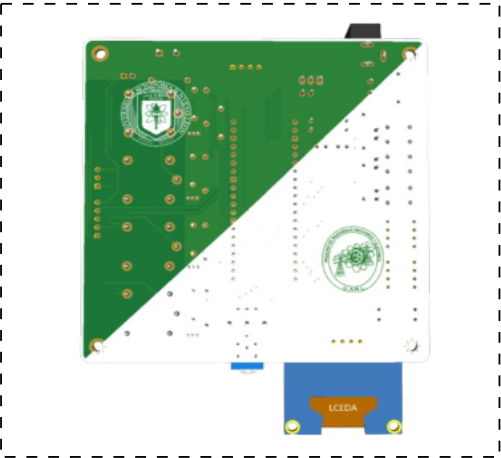
Sistema Embebido RaspBerry

Page	Index	Page	Index	Page	Index
1.	COVER PAGE	4.	PIN OUT	7.	LAYOUT
2.	BLOCK DIAGRAM	5.	MAIN	8.	MECHANICAL
3.	PIN LIST	6.	ORIGINAL	9.	3D MODEL
				10.	REVISION HISTORY

TOP PCB VIEW



BOTTOM PCB VIEW



INFO NOTE:
Example text for a informational design note

CAUTION NOTE:
Example text for a cautionary design note

LAYOUT NOTE:
Example text for a informational layout guidelines note

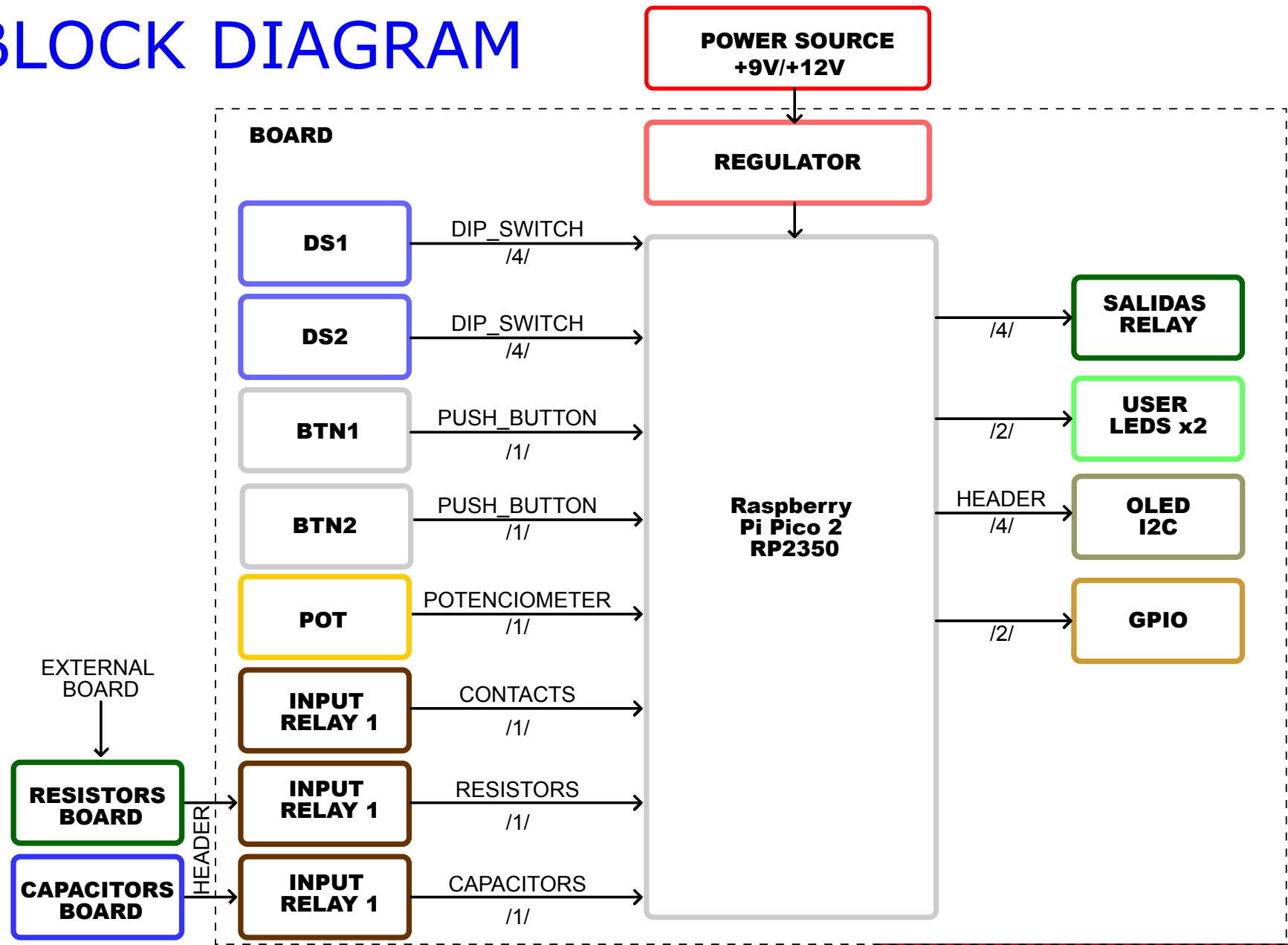
WARNING NOTE:
Example text for a critical design note



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TITLE: COVER PAGE		REV: 1.1
Company: UANL - FIME		Sheet: 1/10
Date: 15/11/25	Drawn By: JesusGCamarena	

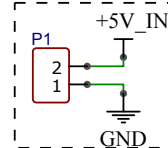
BLOCK DIAGRAM



<https://datasheets.raspberrypi.com/pico/pico-datasheet.pdf>
<https://uelectronics.com/wp-content/uploads/2024/10/Datasheet-Raspberry-Pi-Pico-2.pdf>
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TITLE: BLOCK DIAGRAM		REV: 1.1
Company: UANL - FIME		Sheet: 2/10
Date: 15/11/25		Drawn By: JesusGCamarena

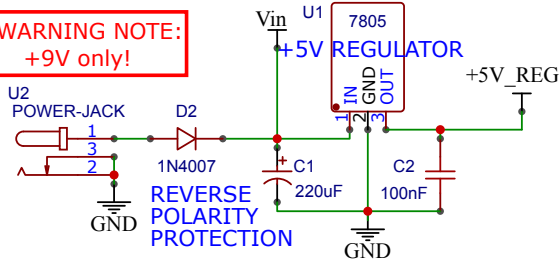
EXTERNAL +5V POWER



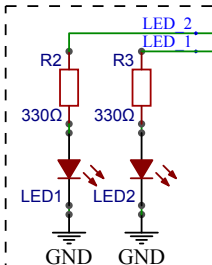
WARNING NOTE:
+5V only!

POWER IN

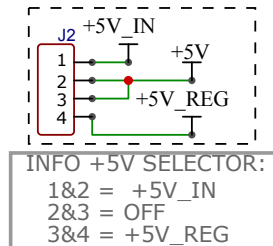
WARNING NOTE:
+9V only!



USER LEDS

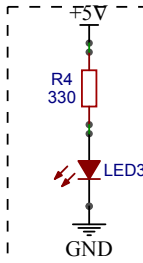


POWER IN SELECTOR

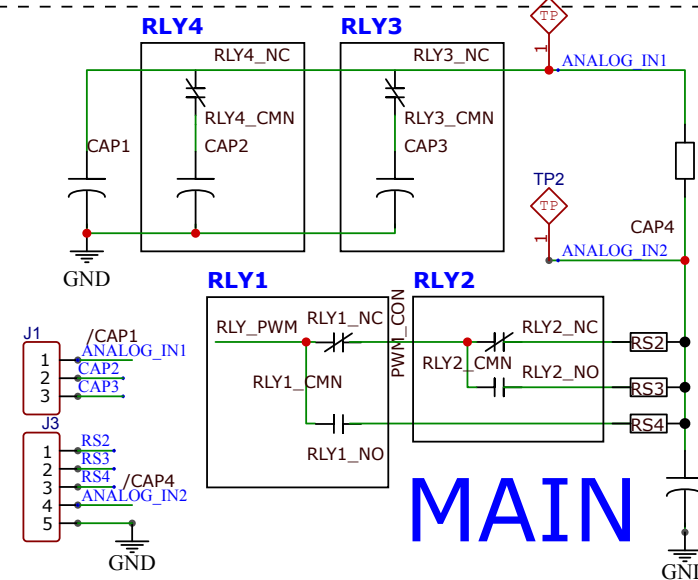


INFO +5V SELECTOR:
1&2 = +5V_IN
2&3 = OFF
3&4 = +5V_REG

POWER ON

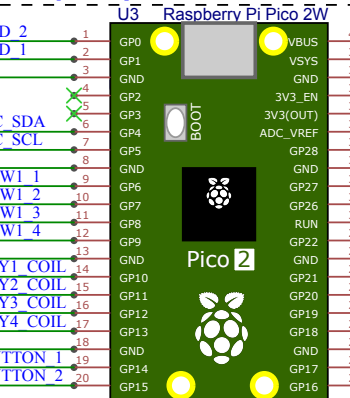


RLY CONECTION

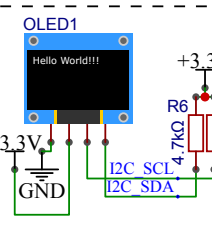


MAIN

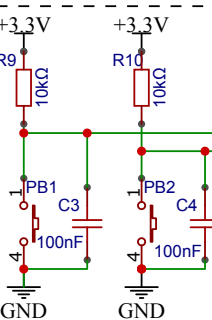
Raspberry Pi Pico 2 RP2350



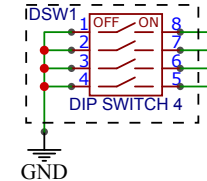
PANTALLA I2C



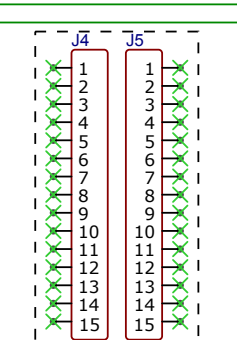
USER BUTTONS



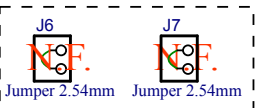
DIP SWITCH 1



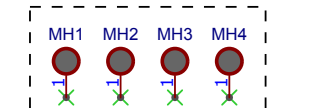
Rasp Pico HEADER



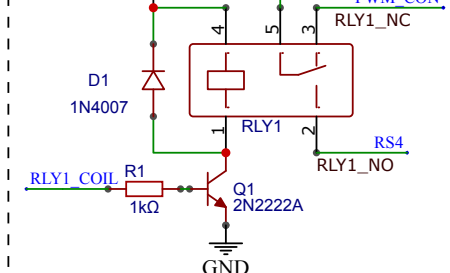
SELECTOR JUMPERS



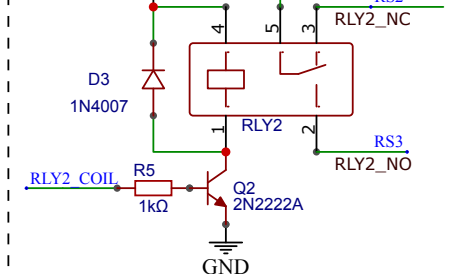
MOUNTING HOLES 3mm



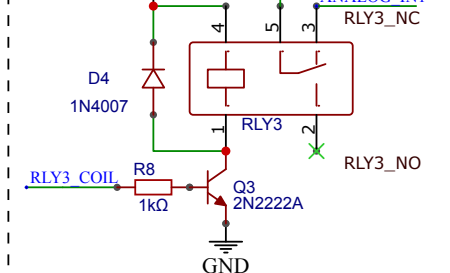
RLY1



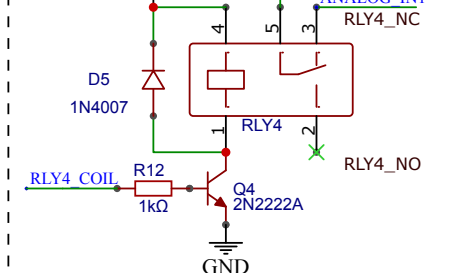
RLY2



RLY3

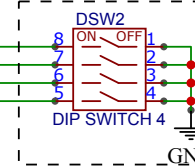


RLY4

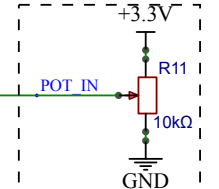


LAYOUT NOTE:
Use when external supply

DIP SWITCH 2



POTENTIOMETER



TITLE:
MAIN

REV: 1.1

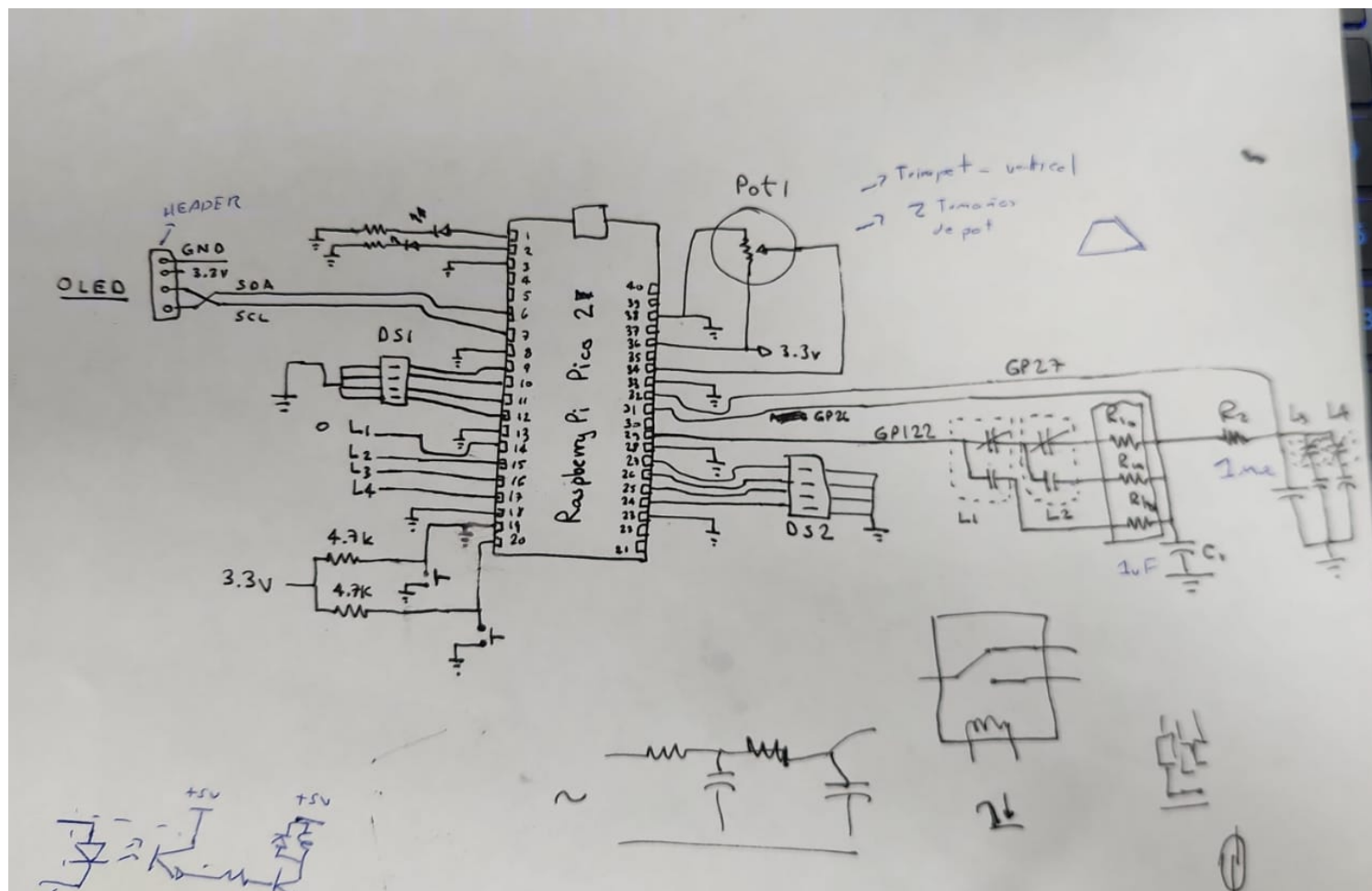
Company: UANL - FIME

Sheet: 5/10

Date: 15/11/25

Drawn By: JesusGCamarena

ORIGINAL LAYOUT



TITLE: Original Layout		REV: 1.1
Company: UANL - FIME		Sheet: 6/10
Date: 15/11/25	Drawn By: JesusGCamarena	

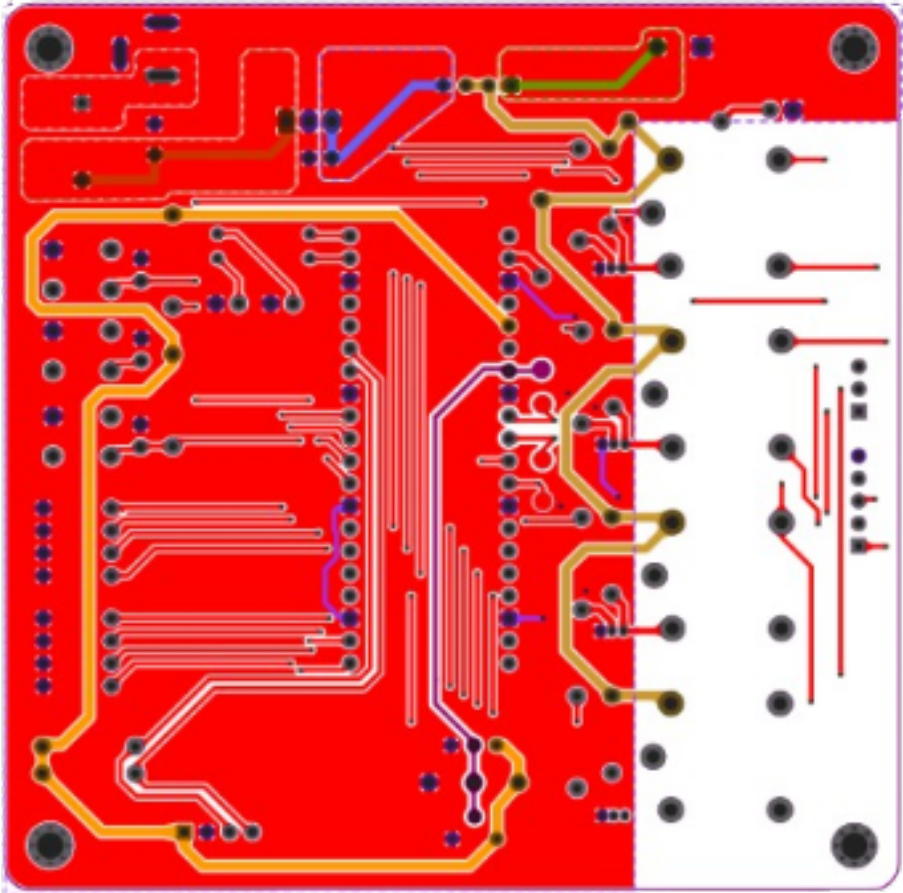
REVISION HISTORY

Nº	Revision/Issue:	Date:
	Initial Design	04/07/25
	Conexiones de la fuente VSYS y VBUS	28/10/25
	Conexiones RLY3 y RLY4	28/10/25
	Pines que soportan +5V	28/10/25
	First Relase	15/11/25

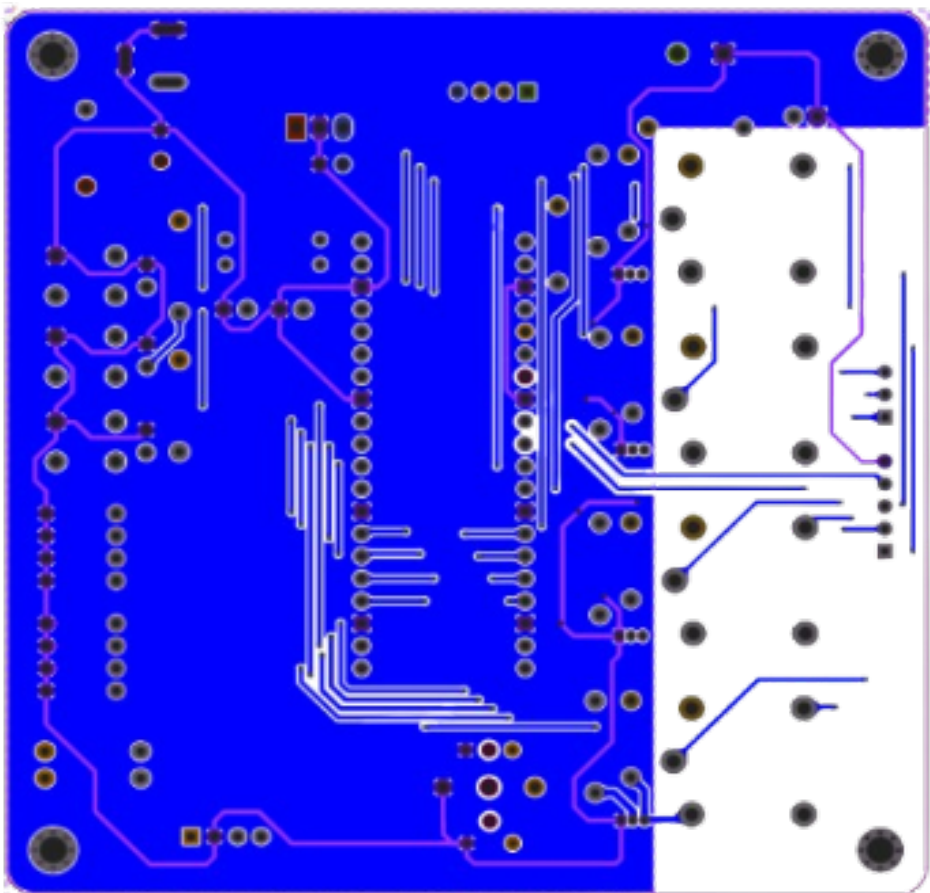
TITLE: REVISION HISTORY		REV: 1.1
	Company: UANL - FIME	Sheet: 10/10
	Date: 15/11/25	Drawn By: JesusGCamarena

LAYOUT

TOP COPPER



BOTTOM VIEW



TITLE: LAYOUT		REV: 1.1
Company: UANL - FIME		Sheet: 7/10
Date: 15/11/25		Drawn By: JesusGCamarena

PIN LIST

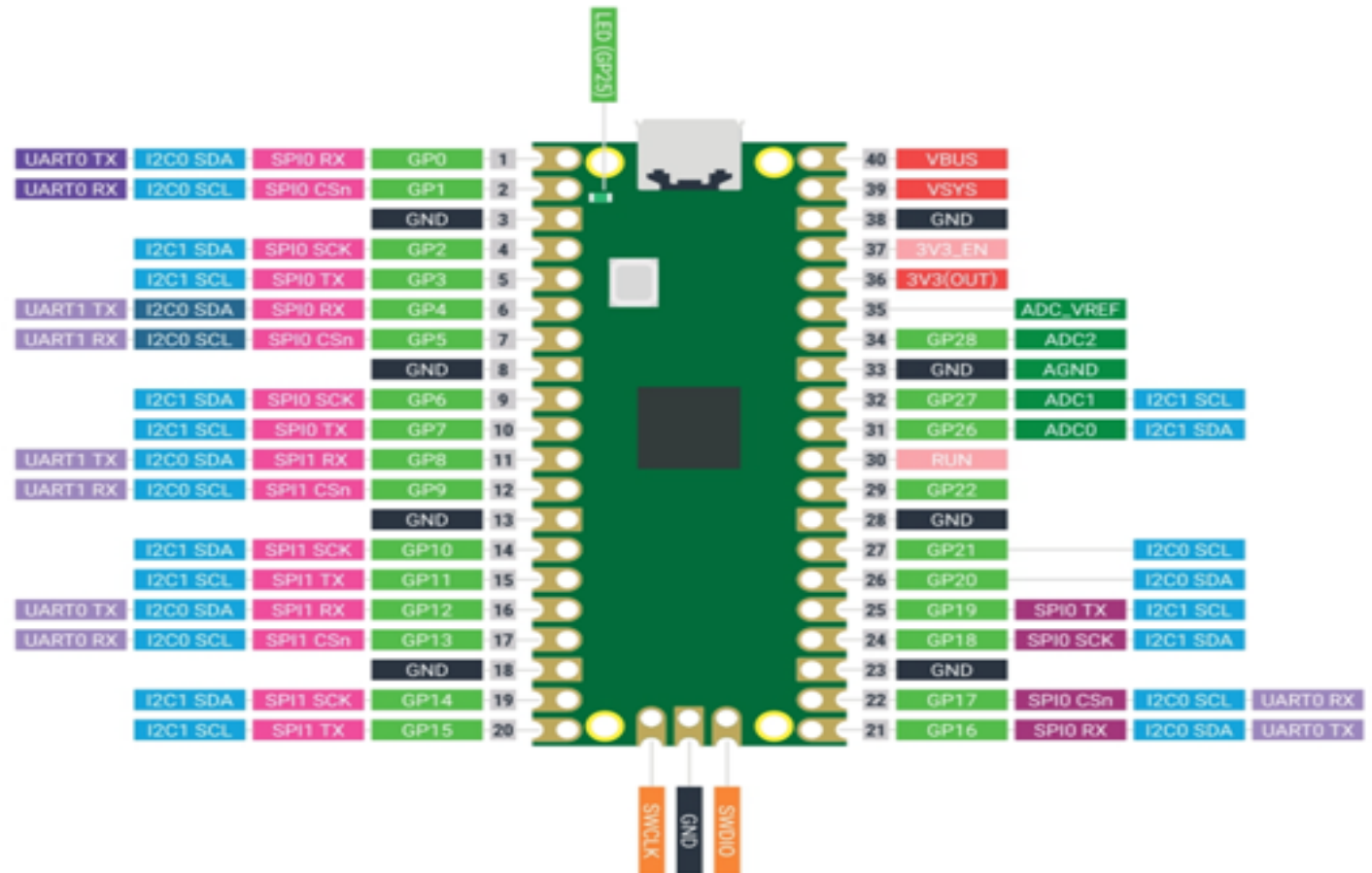
Raspberry Pi Pico												
#	NAME	GPI	RESTRICTIO	TYPE	# ADC	PWM	USART	I2C	SPI	I/O	FUNCTION	NAME
1	GP0	GPIO0	--	GPIO	--	PWM0A	USART0_TX	I2C0_SDA	SPI0_RX	--	--	
2	GP1	GPIO1	--	GPIO	--	PWM0B	USART0_RX	I2C0_SCL	SPI0_CS	--	--	
3	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
4	GP2	GPIO2	--	GPIO	--	PWM1A	--	I2C1_SDA	SPI0_SCK	--	--	
5	GP3	GPIO3	--	GPIO	--	PWM1B	--	I2C1_SCL	SPI0_TX	--	--	
6	GP4	GPIO4	--	GPIO	--	PWM2A	USART1_TX	I2C0_SDA	SPI0_RX	--	--	
7	GP5	GPIO5	--	GPIO	--	PWM2B	USART1_RX	I2C0_SCL	SPI0_CS	--	--	
8	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
9	GP6	GPIO6	--	GPIO	--	PWM3A	--	I2C1_SDA	SPI0_SCK	--	--	
10	GP7	GPIO7	--	GPIO	--	PWM3B	--	I2C1_SCL	SPI0_TX	--	--	
11	GP8	GPIO8	--	GPIO	--	PWM4A	USART1_TX	--	SPI1_RX	--	--	
12	GP9	GPIO9	--	GPIO	--	PWM4B	USART1_RX	--	SPI1_CS	--	--	
13	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
14	GP10	GPIO10	--	GPIO	--	PWM5A	--	--	SPI1_SCK	--	--	
15	GP11	GPIO11	--	GPIO	--	PWM5B	--	--	SPI1_TX	--	--	
16	GP12	GPIO12	--	GPIO	--	PWM6A	USART0_TX	--	SPI1_RX	--	--	
17	GP13	GPIO13	--	GPIO	--	PWM6B	USART0_RX	--	SPI1_CS	--	--	
18	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
19	GP14	GPIO14	--	GPIO	--	PWM7A	--	--	SPI1_SCK	--	--	
20	GP15	GPIO15	--	GPIO	--	PWM7B	--	--	SPI1_TX	--	--	
21	GP16	GPIO16	--	GPIO	--	PWM0A	USART0_TX	--	SPI0_RX	--	--	
22	GP17	GPIO17	--	GPIO	--	PWM0B	USART0_RX	--	SPI0_CS	--	--	
23	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
24	GP18	GPIO18	--	GPIO	--	PWM1A	--	I2C1_SDA	SPI0_SCK	--	--	
25	GP19	GPIO19	--	GPIO	--	PWM1B	--	I2C1_SCL	SPI0_TX	--	--	
26	GP20	GPIO20	LED_BUILTIN	GPIO	--	PWM2A	--	I2C0_SDA	--	--	LED_BUILTIN	
27	GP21	GPIO21	--	GPIO	--	PWM2B	--	I2C0_SCL	--	--	--	
28	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
29	GP22	GPIO22	--	GPIO	--	PWM3A	--	--	--	--	--	
30	RUN	--	RESET	CTRL	--	--	--	--	--	--	RESET	
31	GP26	GPIO26	--	GPIO	ADC0	PWM4A	--	I2C1_SDA	--	--	--	
32	GP27	GPIO27	--	GPIO	ADC1	PWM4B	--	I2C1_SCL	--	--	--	
33	AGND	--	POWER	POWER	--	--	--	--	--	--	POWER	
34	GP28	GPIO28	--	GPIO	ADC2	PWM5A	--	--	--	--	--	
35	ADC_VREF	--	POWER	POWER	--	--	--	--	--	--	POWER	
36	3V3(OUT)	--	POWER	POWER	--	--	--	--	--	--	POWER	
37	3V3_EN	--	CTRL	CTRL	--	--	--	--	--	--	CTRL	
38	GND	--	POWER	POWER	--	--	--	--	--	--	POWER	
39	VSYS	--	POWER	POWER	--	--	--	--	--	--	POWER	
40	VBUS	--	POWER	POWER	--	--	--	--	--	--	POWER	

TITLE: PIN LIST		REV: 1.1
	Company: UANL - FIME	Sheet: 3/10
	Date: 15/11/25	Drawn By: JesusGCamarena

Raspberry Pi Pico Pinout

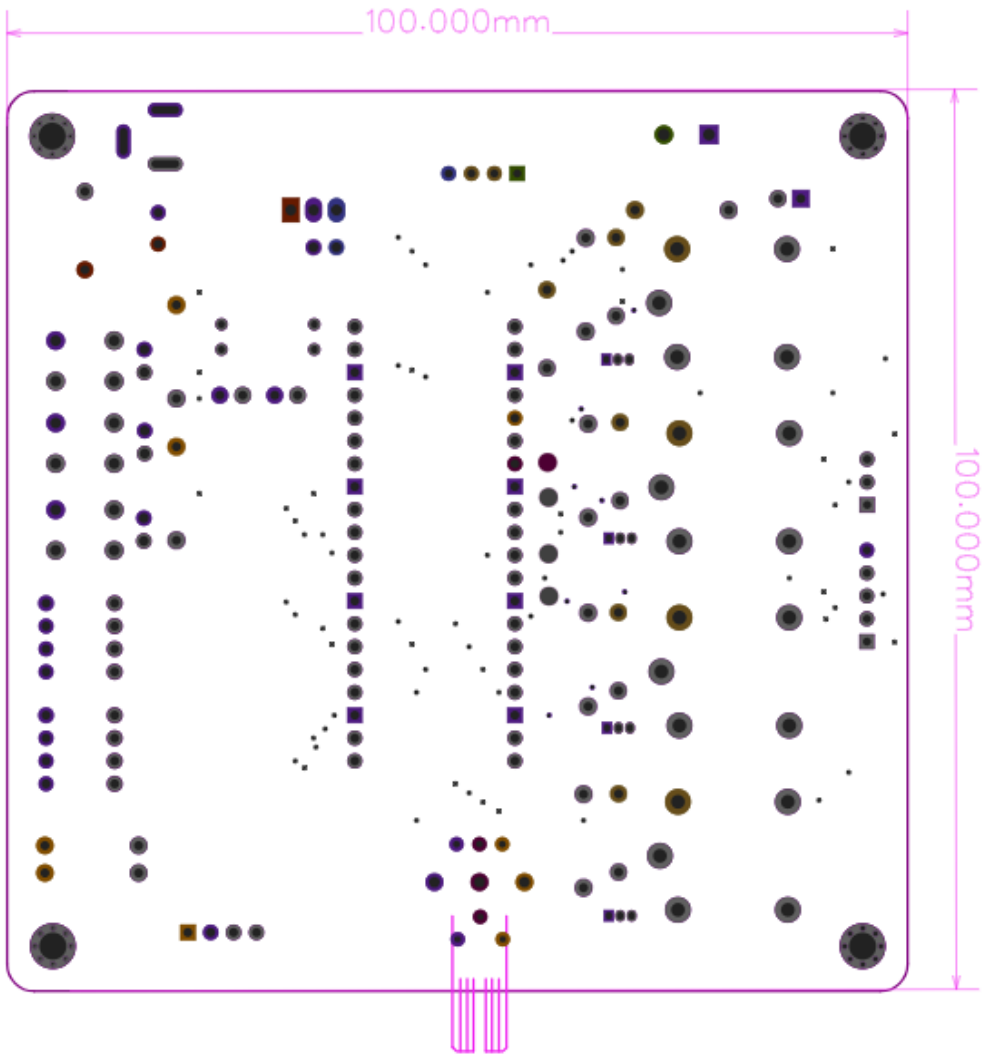
For Raspberry Pi Pico 1 and Raspberry Pi Pico 2

Power	
Ground	
UART / UART (default)	
GPIQ, PIQ, and PWM	
ADC	
SPI / SPI (default)	
I2C / I2C (default)	
System Control	
Debugging	



TITLE: Pin out		REV: 1.1
Company: UANL - FIME		Sheet: 4/10
Date: 15/11/25	Drawn By: JesusGCamarena	

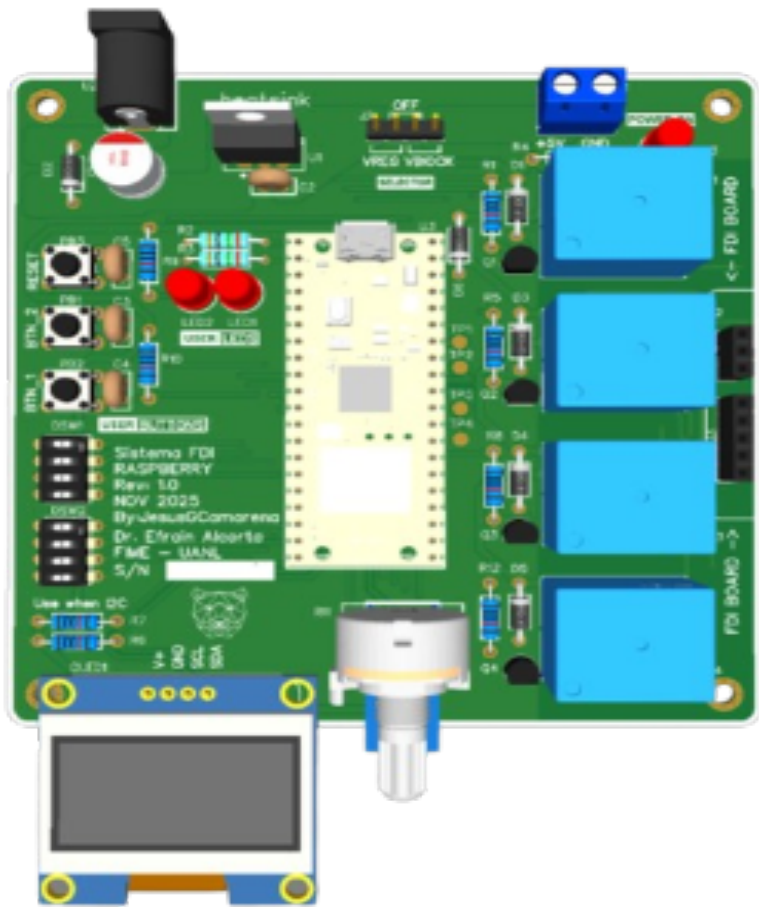
Mechanical



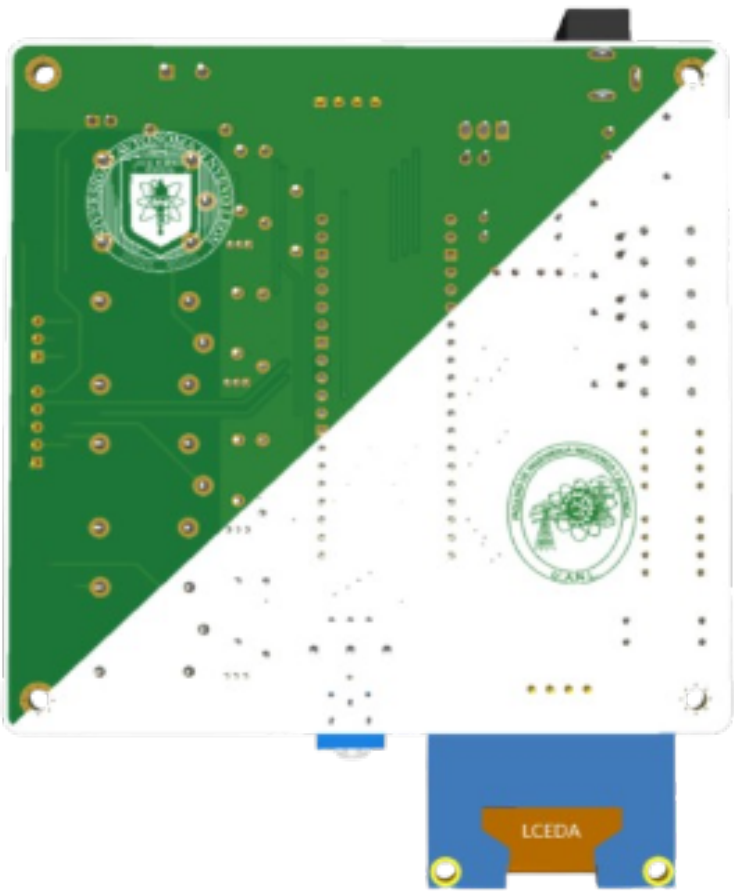
TITLE: MECHANICAL		REV: 1.1
	Company: UANL - FIME	Sheet: 8/10
	Date: 15/11/25	Drawn By: JesusGCamarena

3D MODEL

TOP PCB VIEW



BOTTOM PCB VIEW



TITLE: 3D MODEL		REV: 1.1
Company: UANL - FIME		Sheet: 9/10
Date: 15/11/25		Drawn By: JesusGCamarena